

QDT2

Abstract Algebra

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher who has taught number theory for nine years is preparing an answer key with 40 practice problems. One problem asks students to apply the division algorithm to divide -23 by 6 . Which quotient q and remainder r should appear in the key?

- A. $q = -4, r = 1$
- B. $q = -3, r = -5$
- C. $q = -4, r = -1$
- D. $q = 3, r = 5$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A curriculum writer is building an example of Euclid's algorithm for a unit that runs three weeks. She chooses 252 and 198 because both appear in a tiling puzzle on the unit's cover page. What is $\gcd(252, 198)$?

- A. 18
- B. 36
- C. 6
- D. $2,772$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Two positive integers have a product of $4,320$ and a greatest common divisor of 12 . What is their least common multiple?

- A. $51,840$
- B. 360
- C. 12
- D. $4,320$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

For a cipher club that meets on Thursdays, an advisor builds a multiplicative cipher on the 26-letter alphabet using the key 7. The club printed 50 decoder cards last year. To decode, students need the inverse of 7 modulo 26. What is it?

- A. 11
- B. $1/7$
- C. 15
- D. 19

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which element of Z_{12} has no multiplicative inverse?

- A. 11
- B. None; every nonzero element of Z_{12} has an inverse.
- C. 9
- D. 7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A marching band director with a new budget of \$8,000 lines students up for a parade. In rows of 3, two students are left over; in rows of 5, three are left over. The band won a regional trophy last spring. What is the smallest positive number of students that fits both conditions?

- A. 23
- B. 13
- C. 8
- D. 15

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student in an honors seminar of 14 people claims the nonzero integers form a group under ordinary multiplication. She has already checked that the product of two nonzero integers is a nonzero integer. Which group axiom fails?

- A. Commutativity, which every group must have.
- B. Inverses: most elements, such as 2, have no integer inverse.
- C. Closure: some products leave the set.
- D. Identity: there is no identity element.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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8.

A group G of order 36 has a subgroup H with 9 elements. The textbook, now in its fourth edition, lists G as a symmetry group used in chemistry. How many distinct left cosets of H are there in G ?

- A. 27
- B. 9
- C. It depends on whether G is abelian.
- D. 4

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A finite group G has order 20. Which of these cannot be the order of an element of G ?

- A. 5
- B. 20
- C. 10
- D. 8

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10.

Working a warm-up on a whiteboard that holds about 20 lines, a student computes $4 \cdot 5$ in $\mathbb{Z}_{10}$ and gets 0. The class had just finished a unit on polynomial rings. What does this result show about $\mathbb{Z}_{10}$?

- A. $\mathbb{Z}_{10}$ is not an integral domain, so it is not a field.
- B. $\mathbb{Z}_{10}$ is not a ring at all.
- C. $\mathbb{Z}_{10}$ is a field because it is a finite commutative ring with unity.
- D. $\mathbb{Z}_{10}$ is an integral domain but not a field.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

For which n is $\mathbb{Z}_n$ a field?

- A. All of them, since each $\mathbb{Z}_n$ is a ring with unity.
- B. 9
- C. 15
- D. 11

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A homomorphism ϕ maps a group G of order 30 into a group H of order 60. A graduate assistant grading 25 proofs notes that the kernel of ϕ has 5 elements. How many elements are in the image of ϕ ?

- A. 5
- B. 6
- C. 150
- D. 60, since a homomorphism always maps onto H .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher ~~who has taught number theory for nine years~~ is preparing an answer key ~~with 40 practice problems~~. One problem asks students to apply the division algorithm to divide -23 by 6 . Which quotient q and remainder r should appear in the key?

Target word: q and r for a negative dividend **Decides:** The remainder must satisfy $0 \leq r < 6$.

A	Correct	$6 \times (-4) + 1 = -23$, and $0 \leq 1 < 6$.
B	Dropped element	$6 \times (-3) - 5 = -23$ checks the arithmetic but drops the rule that r cannot be negative.
C	Inverted operation	Sign error: $6 \times (-4) - 1 = -25$, not -23 .
D	Confusion set	That divides $+23$ by 6 ; the dividend is negative.

2. Answer A Concept 2

A curriculum writer is building an example of Euclid's algorithm ~~for a unit that runs three weeks~~. She chooses 252 and 198 ~~because both appear in a tiling puzzle on the unit's cover page~~. What is $\gcd(252, 198)$?

Target word: the greatest common divisor **Decides:** $252 = 198 \cdot 1 + 54$; $198 = 54 \cdot 3 + 36$; $54 = 36 \cdot 1 + 18$; $36 = 18 \cdot 2 + 0$, so the last nonzero remainder is 18 .

A	Correct	The last nonzero remainder in Euclid's algorithm is 18 .
B	Dropped element	Stops one step early; 36 does not divide 54 .
C	Adjacent category	A common divisor, but not the greatest; $252 = 2^2 \cdot 3^2 \cdot 7$ and $198 = 2 \cdot 3^2 \cdot 11$ share $2 \cdot 3^2 = 18$.
D	Confusion set	That is the LCM: $252 \times 198 / 18$.

3. Answer B Concept 2 · transfer

Two positive integers have a product of $4,320$ and a greatest common divisor of 12 . What is their least common multiple?

Target word: solve for the LCM **Decides:** $\gcd \times \text{lcm} = \text{product}$, so $\text{lcm} = 4,320 / 12$.

A	Inverted operation	Multiplies by the GCD instead of dividing.
B	Correct	$4,320 / 12 = 360$.
C	Wrong stem number	That is the GCD given in the stem.
D	Dropped element	Treats the product as the LCM, ignoring the shared factor.

4. Answer C Concept 3

For a cipher club ~~that meets on Thursdays~~, an advisor builds a multiplicative cipher on the 26-letter alphabet using the key 7. ~~The club printed 50 decoder cards last year.~~ To decode, students need the inverse of 7 modulo 26. What is it?

Target word: multiplicative inverse mod 26 **Decides:** Find x with $7x \equiv 1 \pmod{26}$; $7 \times 15 = 105 = 4 \cdot 26 + 1$.

A	Inverted operation	$7 \times 11 = 77 \equiv -1 \pmod{26}$, the negative of what is needed.
B	Adjacent category	A rational number is not an element of $\mathbb{Z}_{26}$; the inverse must be an integer residue.
C	Correct	$7 \times 15 = 105 \equiv 1 \pmod{26}$.
D	Confusion set	$26 - 7 = 19$ is the additive inverse: $7 + 19 \equiv 0$.

5. Answer C Concept 3 • transfer

Which element of $\mathbb{Z}_{12}$ has no multiplicative inverse?

Target word: find the element that fails — reverse of computing an inverse **Decides:** An inverse exists only when $\gcd(a, 12) = 1$; $\gcd(9, 12) = 3$.

A	Confusion set	$11 \times 11 = 121 \equiv 1 \pmod{12}$; 11 is its own inverse.
B	Absolute language	That holds only when the modulus is prime.
C	Correct	$\gcd(9, 12) = 3 \neq 1$, so no inverse exists.
D	Adjacent category	$7 \times 7 = 49 \equiv 1 \pmod{12}$; being odd or prime is not the test, coprimality is.

6. Answer C Concept 4

A marching band director ~~with a new budget of \$8,000~~ lines students up for a parade. In rows of 3, two students are left over; in rows of 5, three are left over. ~~The band won a regional trophy last spring.~~ What is the smallest positive number of students that fits both conditions?

Target word: smallest solution of a congruence system **Decides:** $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$; the unique solution mod 15 is 8.

A	Adjacent category	23 is a solution, but it is $8 + 15$, not the smallest.
B	Dropped element	$13 \equiv 3 \pmod{5}$ but $13 \equiv 1 \pmod{3}$; it meets only one condition.
C	Correct	$8 = 2 \cdot 3 + 2$ and $8 = 1 \cdot 5 + 3$; it is the least positive solution mod 15.
D	Confusion set	15 is the combined modulus, not a solution; it leaves remainder 0 in both.

7. Answer B Concept 5

A student ~~in an honors seminar of 14 people~~ claims the nonzero integers form a group under ordinary multiplication. She has already checked that the product of two nonzero integers is a nonzero integer. Which group axiom fails?

Target word: which axiom fails **Decides:** 2 has no integer x with $2x = 1$.

A	Absolute language	Groups need not be commutative, and multiplication here is commutative anyway.
B	Correct	$1/2$ is not an integer, so the inverse axiom fails.
C	Confusion set	The stem already shows closure holds.
D	Adjacent category	1 is the multiplicative identity and is in the set.

8. Answer D Concept 6

A group G of order 36 has a subgroup H with 9 elements. The textbook, ~~now in its fourth edition,~~ lists G as a symmetry group ~~used in chemistry.~~ How many distinct left cosets of H are there in G ?

Target word: number of cosets (the index) **Decides:** Index = $|G| / |H| = 36 / 9$.

A	Inverted operation	Subtracts the orders instead of dividing.
B	Wrong stem number	That is the size of each coset, not how many there are.
C	Adjacent category	Commutativity matters for whether H is normal, not for how many cosets exist.
D	Correct	$36 / 9 = 4$ cosets, each with 9 elements.

9. Answer D Concept 6 · transfer

A finite group G has order 20. Which of these cannot be the order of an element of G ?

Target word: an impossible order — reverse use of Lagrange **Decides:** Element orders must divide 20.

A	Confusion set	5 divides 20; by Cauchy's theorem an element of order 5 must exist.
B	Absolute language	An element can have order $ G $ when G is cyclic.
C	Confusion set	10 divides 20, so it is possible.
D	Correct	8 does not divide 20.

10. Answer A Concept 7

Working a warm-up ~~on a whiteboard that holds about 20 lines,~~ a student computes $4 \cdot 5$ in $\mathbb{Z}_{10}$ and gets 0. ~~The class had just finished a unit on polynomial rings.~~ What does this result show about $\mathbb{Z}_{10}$?

Target word: classify $\mathbb{Z}_{10}$ **Decides:** Two nonzero elements multiply to 0, so $\mathbb{Z}_{10}$ has zero divisors.

A	Correct	Zero divisors rule out an integral domain, and every field is an integral domain.
B	Confusion set	$\mathbb{Z}_{10}$ is a commutative ring with unity; zero divisors do not break the ring axioms.
C	Dropped element	Drops the condition of no zero divisors; only finite integral domains are fields.
D	Adjacent category	An integral domain cannot have zero divisors.

11. Answer D Concept 7 · transfer

For which n is $\mathbb{Z}_n$ a field?

Target word: choose n — reverse from property to modulus **Decides:** $\mathbb{Z}_n$ is a field exactly when n is prime.

A	Absolute language	Being a ring with unity is not enough; composite n gives zero divisors.
B	Confusion set	Odd but not prime: $3 \cdot 3 \equiv 0 \pmod{9}$.
C	Confusion set	$3 \cdot 5 \equiv 0 \pmod{15}$, a zero divisor.
D	Correct	11 is prime, so every nonzero element has an inverse.

12. Answer B Concept 8

A homomorphism ϕ maps a group G of order 30 into a group H of order 60. A graduate assistant ~~grading~~^{proofs} notes that the kernel of ϕ has 5 elements. How many elements are in the image of ϕ ?

Target word: size of the image **Decides:** $|\text{image}| = |G| / |\ker \phi| = 30 / 5$.

A	Wrong stem number	That is the kernel size.
B	Correct	By the first isomorphism theorem, $G / \ker \phi \cong \text{image}$, so $30 / 5 = 6$.
C	Inverted operation	Multiplies the orders instead of dividing.
D	Absolute language	Homomorphisms need not be onto.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The division algorithm	For integers a and $b > 0$ there are unique q and r with $a = bq + r$ and $0 \leq r < b$; the remainder is never negative.	Giving a negative remainder when a is negative.	A negative dividend and a request for q and r .
2	Euclid's algorithm, GCD and LCM	The GCD is the last nonzero remainder in Euclid's algorithm, and $\gcd(a, b) \times \text{lcm}(a, b) = a \times b$ for positive integers.	Stopping a step early, or confusing the GCD with the LCM.	Two integers and the words "greatest common divisor" or "least common multiple."
3	Multiplicative inverses mod n	a has an inverse mod n exactly when $\gcd(a, n) = 1$; the inverse x satisfies $ax \equiv 1 \pmod{n}$.	Using the additive inverse $n - a$, or assuming every nonzero element has an inverse.	"Inverse of a modulo n " or "solve $ax \equiv 1$."
4	Chinese remainder theorem	If the moduli are pairwise coprime, a system of congruences has exactly one solution modulo their product.	Accepting a number that satisfies only one of the congruences.	Two or more "leaves a remainder of" conditions with different moduli.
5	Group axioms	A group needs closure, associativity, an identity, and an inverse for every element; commutativity is not required.	Calling (nonzero integers, $\times$) a group, or requiring commutativity.	"Is this set a group under this operation?"
6	Lagrange's theorem and cosets	In a finite group, the order of every subgroup and every element divides $ G $, and the number of cosets is $ G / H $.	Subtracting orders, or thinking any number up to $ G $ can be an order.	A group order and a subgroup or element order.
7	Integral domains and fields	$\mathbb{Z}_n$ is an integral domain, and in fact a field, exactly when n is prime; a composite n gives zero divisors.	Treating every finite commutative ring with unity as a field.	A product of two nonzero elements that equals zero, or "for which n is $\mathbb{Z}_n$ a field?"
8	Homomorphisms and kernels	For a homomorphism $\phi: G \rightarrow H$, $ \text{image} = G / \text{kernel} $ (first isomorphism theorem).	Assuming a homomorphism is onto, or multiplying instead of dividing.	A group order and a kernel order, asking about the image.

Course scope (public description): Abstract Algebra is the axiomatic and rigorous study of the underlying structure of algebra and arithmetic. It covers the knowledge and skills necessary to understand, apply, and prove theorems about numbers, groups, rings, and fields. Topics include the well-ordering principle, equivalence classes, the division algorithm, Euclid's algorithm, prime factorization, greatest common divisor, least common multiple, congruence, the Chinese remainder theorem, modular arithmetic, rings, integral domains, fields, groups, roots of unity, and homomorphisms. Linear Algebra is a prerequisite for this course.